MAT 3007 – Optimization

# KKT Condition

*Lecture 19*

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# Recap

# Abstract FONC for Constrained Problems

## Theorem: FONC for Constrained Problems

Let $\boldsymbol{x}^*$ be a local minimizer of $\min_{\boldsymbol{x}\in\Omega} f(\boldsymbol{x})$. Then for any feasible direction $\boldsymbol{d}$ at $\boldsymbol{x}^*$, we must have $\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0$.

# Abstract FONC for Constrained Problems

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**Remark:**

- feasible direction: $\boldsymbol{d}$ a feasible direction at $\boldsymbol{x} \in \Omega$ if $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all $t$ small enough.
- descent direction: $\boldsymbol{d}$ is a descent direction at $\boldsymbol{x}$ if and only if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.
- Therefore, this abstract FONC says that there are no feasible descent directions.

# Problems with Linear Inequality Constraints

Consider an inequality constrained problem:

$$\text{minimize}_{\boldsymbol{x}} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}. \tag{1}$$

## Theorem: FONC for Linearly Constrained Problems

If $\boldsymbol{x}^*$ is a local minimizer of (1), then there exists some $\boldsymbol{y} \geq \boldsymbol{0}$ with

$$\nabla f(\boldsymbol{x}^*) + \boldsymbol{A}^\top \boldsymbol{y} = \boldsymbol{0}$$
$$y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x}^* - b_i) = 0 \quad \forall \; i,$$

where $\boldsymbol{a}_i^\top$ is the $i$th row of $\boldsymbol{A}$.

- This is a special case of KKT condition with linear constraints.

# Abstract FONC for Inequality Constrained Prob.

Consider the problem with nonlinear inequality constraints:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \leq 0, \quad \forall\ i = 1, \ldots, m. \end{aligned} \tag{2}$$

- $f$ and $g_i$, $i = 1, \ldots, m$, are assumed to be continuously differentiable

## Lemma: No Feasible Descent for Inequality Constraints

Let $\boldsymbol{x}^*$ be a local minimizer of (2). Then, there does not exist a vector $\boldsymbol{d} \in \mathbb{R}^n$ such that:

$$\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \text{and} \quad \nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \forall i \in \mathcal{A}(\boldsymbol{x}^*).$$

descent direction

feasible direction

set of indexes for active constraints.

## Theorem: Fritz-John Conditions

Let $\boldsymbol{x}^*$ be a local minimizer of (2). Then, there exists $\lambda_0 \geq 0$, $\lambda_1 \geq 0$, ..., $\lambda_m \geq 0$, which are not all zeros, such that

$$\lambda_0 \nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0,$$

$$\lambda_i \cdot g_i(\boldsymbol{x}^*) = 0, \quad \forall\ i = 1, ..., m.$$

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$$\lambda_i \cdot g_i(\boldsymbol{x}^*) = 0, \quad \forall\ i = 1, ..., m.$$ complementarity.

**Remark:**

- It is a necessary condition.
- However, it allows $\lambda_0 = 0$, which means that no information about $f$ is included. This can lead to inclusion of many points that are not necessarily minimizers.

# KKT Conditions for Inequality Constraints

Recall the nonlinear optimization:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \leq 0, \quad \forall\ i = 1, ..., m. \end{aligned} \tag{3}$$

# KKT Conditions for Inequality Constraints

Recall the nonlinear optimization:

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## Theorem: KKT Conditions for Inequality Constrained Prob.

Let $\boldsymbol{x}^*$ be a local minimizer of (3) and suppose that the vectors

$$\{\nabla g_i(\boldsymbol{x}^*) : i \in \mathcal{A}(\boldsymbol{x}^*)\}$$

are linearly independent. Then, there exist $\lambda_1, ..., \lambda_m \ge 0$ such that

$$\begin{aligned} \nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0, \\ \lambda_i \cdot g_i(\boldsymbol{x}^*) = 0, \quad i = 1, ..., m. \end{aligned}$$

# Proof

By Fritz-John Conditions, assume that $\lambda_0 = 0$, then

$$\sum_{i \in \mathcal{A}(\boldsymbol{x}^*)} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0.$$

Since $\{\nabla g_i(\boldsymbol{x}^*) : i \in \mathcal{A}(\boldsymbol{x}^*)\}$ are linearly independent, then $\lambda_i = 0$ for $i \in \mathcal{A}(\boldsymbol{x}^*)$. Also, $\lambda_i = 0$ for $i \in \mathcal{I}(\boldsymbol{x}^*)$. Therefore, $\lambda_0, \lambda_1, \ldots, \lambda_m = 0$, contradicting the fact that $\lambda_0, \lambda_1, \ldots, \lambda_m$ are not all zeros in FJ conditions.

Then, we conclude that $\lambda_0 > 0$, dividing two sides by $\lambda_0$ gives the results.

# KKT Conditions with Inequality and Equality Constraints

# KKT Conditions: The General Setting

## General Nonlinear Optimization Problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\ i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1, \ldots, p. \end{aligned}$$

- We have already derived the KKT conditions for problems with linear constraint or nonlinear inequality constraints.

$\rightsquigarrow$ Let's develop the general KKT conditions from another perspective.

# The Lagrangian

We define the Lagrangian of this problem by:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j h_j(\boldsymbol{x})$$

where we associate each constraint with a Lagrangian multiplier:

$$\begin{aligned} g_i(\boldsymbol{x}) \leq 0 \quad &\rightarrow \quad \lambda_i \geq 0 \\ h_j(\boldsymbol{x}) = 0 \quad &\rightarrow \quad \mu_j \end{aligned}$$

can also be called dual variables.

**Lagrangian Multipliers**:

- The multipliers for inequality constraints $g_i(\boldsymbol{x}) \leq 0$ are non-negative: $\lambda_i \geq 0$.
- The multipliers for the equality constraints are free: $\mu_j \in \mathbb{R}$.
- The condition "$\lambda_i \geq 0$, $\mu_i$ free" are called dual feasibility.

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**Meaning of the Lagrangian**:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j h_j(\boldsymbol{x})$$

$\rightsquigarrow$ We move all the constraints to the objective and interpret them as reward/penalty functions.

- We want $g_i(\boldsymbol{x}) \leq 0$: setting $\lambda_i \geq 0$, we reduce the Lagrangian when the constraint holds and penalize/increase it when the constraint does not hold.

# Derive KKT from Lagrangian

Take derivative with respect to $\boldsymbol{x}$ and set it to zero.

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = 0.$$

$$\nabla_x L(x, \lambda, \mu) = \nabla f(x) + \sum \lambda_i \nabla g_i(x) + \sum_j \mu_j \nabla h_j(x)$$

- This forms one part of the KKT conditions, which we call the main conditions.

# Derive KKT from Lagrangian

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We also have a set of complementarity conditions:

- Each multiplier is complementary to the constraints it is associated to, i.e.,

$$\begin{aligned} \lambda_i \cdot g_i(\boldsymbol{x}) &= 0, \quad \forall\, i \\ \mu_j \cdot h_j(\boldsymbol{x}) &= 0, \quad \forall\, j. \end{aligned}$$

  The second condition is usually omitted since the feasibility of $\boldsymbol{x}$ already guarantees it.

# Theorem: KKT Conditions

If $\boldsymbol{x}$ is a local minimizer and if a constraint qualification holds, then there exist $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$ such that:

**1. Main Condition**

$$\nabla f(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j \nabla h_j(\boldsymbol{x}) = \mathbf{0}.$$

**2. Dual Feasibility**

$$\lambda_i \geq 0 \quad i = 1, \ldots, m.$$

**3. Complementarity**

$$\lambda_i \cdot g_i(\boldsymbol{x}) = 0 \quad \forall\ i = 1, \ldots, m.$$

**4. Primal Feasibility**

$$g_i(\boldsymbol{x}) \leq 0, \quad h_j(\boldsymbol{x}) = 0 \quad \forall\ i, \quad \forall\ j.$$

# Constraint Qualifications

We require the collection of gradients

$$\{\nabla g_i(\boldsymbol{x}) : i \in \mathcal{A}(\boldsymbol{x})\} \cup \{\nabla h_j(\boldsymbol{x}) : j = 1, \ldots, p\} \qquad (\star)$$

to be linearly independent.

- This condition is a constraint qualification (CQ) called the Linear Independence Constraint Qualification (LICQ).

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- This condition is a constraint qualification (CQ) called the Linear Independence Constraint Qualification (LICQ).
- A feasible point $\boldsymbol{x}$ satisfying LICQ is called regular.
- LICQ ensures that the Karush-Kuhn-Tucker (KKT) conditions are necessary for optimality.
- Other CQs: ACQ, GCQ, MFCQ, PLICQ, Slater's condition

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- A feasible point $\boldsymbol{x}$ satisfying LICQ is called regular.
- LICQ ensures that the Karush-Kuhn-Tucker (KKT) conditions are necessary for optimality.
- Other CQs: ACQ, GCQ, MFCQ, PLICQ, Slater's condition
- For linear constraints: No additional condition is required for the KKT conditions to be necessary.

**Further Remarks:**

- A (feasible) point satisfying the KKT conditions is called a KKT point.
- The main condition can be compactly represented as:

$$\nabla_x L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \nabla f(\boldsymbol{x}) + \nabla g(\boldsymbol{x})\boldsymbol{\lambda} + \nabla h(\boldsymbol{x})\boldsymbol{\mu} = \mathbf{0},$$

where we set

consider $g$ and $h$ vector-value function

$\nabla g(\boldsymbol{x}) := (\nabla g_1(\boldsymbol{x}), \ldots, \nabla g_m(\boldsymbol{x})) = Dg(\boldsymbol{x})^\top \in \mathbb{R}^{m\times n}$ and
$\nabla h(\boldsymbol{x}) := (\nabla h_1(\boldsymbol{x}), \ldots, \nabla h_p(\boldsymbol{x})) = Dh(\boldsymbol{x})^\top \in \mathbb{R}^{p\times n}$.

$$\nabla g(x) = \begin{bmatrix} | & & | \\ \nabla g_1(x) & \cdots & \nabla g_m(x) \\ | & & | \end{bmatrix} \qquad \nabla g(x)\lambda = \begin{bmatrix} | & & | \\ \nabla g_1(x) & \cdots & \nabla g_m(x) \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{bmatrix} = \sum \lambda_i \nabla g_i(x)$$

# Examples: Formulating and Using KKT Conditions

# Example I

Consider the problem:

$$\begin{aligned} \text{minimize} \quad & 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ \text{subject to} \quad & x_1^2 + x_2^2 \leq 5 \quad g_1(x) = x_1^2 + x_2^2 - 5 \leq 0 \\ & 3x_1 + x_2 \geq 3 \quad g_2(x) = -3x_1 - x_2 + 3 \leq 0 \end{aligned}$$

$\rightsquigarrow$ Formulate the KKT conditions.

# Example I

Consider the problem:

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$\rightsquigarrow$ Formulate the KKT conditions.

First we associate the constraints with Lagrange multipliers $\lambda_1$ and $\lambda_2$ and construct the Lagrangian for this problem:

$$\begin{aligned} L(\boldsymbol{x}, \boldsymbol{\lambda}) = & 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ & + \lambda_1(x_1^2 + x_2^2 - 5) + \lambda_2(3 - 3x_1 - x_2) \end{aligned}$$

with $\lambda_1 \geq 0$, $\lambda_2 \geq 0$.

# Example I: Continued

The gradients are given by:

$$\nabla f(\boldsymbol{x}) = \begin{pmatrix} 4x_1 + 2x_2 - 10 \\ 2x_1 + 2_x 2 - 10 \end{pmatrix}, \nabla g_1(\boldsymbol{x}) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}, \nabla g_2(\boldsymbol{x}) = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$$

The main conditions are:

$$\begin{aligned} 4x_1 + 2x_2 - 10 + 2\lambda_1 x_1 - 3\lambda_2 &= 0 \\ 2x_1 + 2x_2 - 10 + 2\lambda_1 x_2 - \lambda_2 &= 0 \end{aligned}$$

The complementarity conditions are:

$$\begin{aligned} \lambda_1 \cdot (x_1^2 + x_2^2 - 5) &= 0 \\ \lambda_2 \cdot (3 - 3x_1 - x_2) &= 0 \end{aligned}$$

# Example I: Continued

Therefore, the KKT conditions are given by:

1. Main Conditions:

$$4x_1 + 2x_2 - 10 + 2\lambda_1 x_1 - 3\lambda_2 = 0$$
$$2x_1 + 2x_2 - 10 + 2\lambda_1 x_2 - \lambda_2 = 0$$

2. Primal Feasibility

$$x_1^2 + x_2^2 \leq 5, \quad -3x_1 - x_2 \leq -3$$

3. Dual Feasibility

$$\lambda_1 \geq 0, \quad \lambda_2 \geq 0$$

4. Complementarity Conditions

$$\lambda_1 \cdot (x_1^2 + x_2^2 - 5) = 0$$
$$\lambda_2 \cdot (3 - 3x_1 - x_2) = 0$$

# Example II

Formulate the KKT conditions for the problem (assume $\boldsymbol{Q}$ is symmetric):

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \tfrac{1}{2}\boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x} - \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \quad \leftarrow \mu \\ & \boldsymbol{C}\boldsymbol{x} \leq \boldsymbol{d} \quad \leftarrow \lambda \\ & \boldsymbol{x} \geq \boldsymbol{0} \quad \leftarrow \eta \end{aligned}$$

- We associate the linear constraints with the multiplier $\boldsymbol{\mu}$.
- We introduce the multipliers $\boldsymbol{\lambda}$ and $\boldsymbol{\eta}$ for the constraints.

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- We associate the linear constraints with the multiplier $\boldsymbol{\mu}$.
- We introduce the multipliers $\boldsymbol{\lambda}$ and $\boldsymbol{\eta}$ for the constraints.

The Lagrangian for this problem is:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}, \boldsymbol{\eta}) = \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x} - \boldsymbol{c}^\top \boldsymbol{x} + \boldsymbol{\mu}^\top(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}) + \boldsymbol{\lambda}^\top(\boldsymbol{C}\boldsymbol{x} - \boldsymbol{d}) - \boldsymbol{\eta}^\top \boldsymbol{x}.$$

We have $\boldsymbol{\lambda}, \boldsymbol{\eta} \geq \boldsymbol{0}$ and $\boldsymbol{\mu}$ is a free variable.

# Example II — Continued

The main condition is:

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}, \boldsymbol{\eta}) = \boldsymbol{Q}\boldsymbol{x} - \boldsymbol{c} + \boldsymbol{A}^\top \boldsymbol{\mu} + \boldsymbol{C}^\top \boldsymbol{\lambda} - \boldsymbol{\eta} = 0.$$

The complementarity conditions are:

$$\begin{aligned} \lambda_i \cdot (\boldsymbol{c}_i^\top \boldsymbol{x} - d_i) = 0, \qquad & \forall\ i \\ -\eta_i \cdot x_i = 0, \qquad & \forall\ i, \end{aligned}$$

where $\boldsymbol{c}_i^\top$ is the $i$th row of $\boldsymbol{C}$.

# Example II — Continued

Therefore, the KKT conditions are:

1. Main condition:

$$\nabla_x L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}, \boldsymbol{\eta}) = \boldsymbol{Q}\boldsymbol{x} - \boldsymbol{c} + \boldsymbol{A}^\top \boldsymbol{\mu} + \boldsymbol{C}^\top \boldsymbol{\lambda} - \boldsymbol{\eta} = \boldsymbol{0}.$$

2. Primal feasibility

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{C}\boldsymbol{x} \leq \boldsymbol{d}, \quad \boldsymbol{x} \geq \boldsymbol{0}.$$

3. Dual feasibility

$$\boldsymbol{\lambda} \geq \boldsymbol{0}, \quad \boldsymbol{\eta} \geq \boldsymbol{0}.$$

4. Complementarity conditions

$$\begin{aligned} \lambda_i \cdot (\boldsymbol{c}_i^\top \boldsymbol{x} - d_i) = 0, \qquad & \forall\ i \\ -\eta_i \cdot x_i = 0, \qquad & \forall\ i. \end{aligned}$$

# Example III: Cylinder Volume

We want to build a cylinder with the maximum volume, with its surface area no larger than $C$.

- Decision variables: $r$ (the radius of the base) and $h$ (height).




$$\begin{aligned} \text{maximize}_{r,h} \quad & \pi r^2 h \\ \text{subject to} \quad & 2\pi r^2 + 2\pi rh \le C \\ & r, h \ge 0 \end{aligned}$$

volumn

surface area

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- Denote $\boldsymbol{x} = (r; h)$. The constraints are

$$\begin{aligned} g_1(\boldsymbol{x}) &= 2\pi r(r+h) - C \leq 0, \\ g_2(\boldsymbol{x}) &= -r \leq 0, \\ g_3(\boldsymbol{x}) &= -h \leq 0. \end{aligned}$$

# Example III: Continued

Convert it to a minimization problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}=(r,h)} \quad & -\pi r^2 h \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad i = 1, 2, 3. \end{aligned}$$

Introduce Lagrangian multiplier $\boldsymbol{\lambda} \ge \mathbf{0}$. The Lagrangian is:

$$L(r, h, \boldsymbol{\lambda}) = -\pi r^2 h + \boldsymbol{\lambda}^\top g(\boldsymbol{x})$$

We have

$$\nabla_{\boldsymbol{x}} L(r, h, \boldsymbol{\lambda}) = \begin{pmatrix} -2\pi rh \\ -\pi r^2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4\pi r + 2\pi h \\ 2\pi r \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

# Example III — KKT Conditions

1. Main condition:

$$-2\pi rh + 4\pi r\lambda_1 + 2\pi h\lambda_1 - \lambda_2 = 0, \quad -\pi r^2 + 2\pi r\lambda_1 - \lambda_3 = 0$$

2. Dual feasibility: $\boldsymbol{\lambda} \geq \mathbf{0}$
3. Complementarity conditions:

$$\begin{aligned} \lambda_1 \cdot (2\pi r^2 + 2\pi rh - C) &= 0 \\ \lambda_2 \cdot r &= 0 \\ \lambda_3 \cdot h &= 0 \end{aligned}$$

4. Primal feasibility:

$$2\pi r^2 + 2\pi rh \leq C, \quad r \geq 0, \quad h \geq 0$$

# Example III: Continued

**Case 1**: $r = 0$. Then, the main condition reduces to

$$2\pi h\lambda_1 - \lambda_2 = 0, \quad \lambda_3 = 0.$$

Since $r = 0$, we have $g_1(x) = -C < 0$. By the first complem. condition, we must have: $\lambda_1 = 0$. Therefore, $\boldsymbol{\lambda} = \mathbf{0}$.

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- We plug in $r = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$ back into KKT conditions. It is easy to conclude that all $(0; h)$ with $h \geq 0$ are KKT points.

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- We plug in $r = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$ back into KKT conditions. It is easy to conclude that all $(0; h)$ with $h \geq 0$ are KKT points.

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- We plug in $r = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$ back into KKT conditions. It is easy to conclude that all $(0; h)$ with $h \geq 0$ are KKT points.

**Case 2**: $r > 0$. By second complem. condition, we have $\lambda_2 = 0$.
**Subcase 2.1**: $\lambda_1 = 0$. By the second main condition, we have $\lambda_3 = -\pi r^2 < 0$, so there is no such KKT points.

# Example III: Continued

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Since $r = 0$, we have $g_1(x) = -C < 0$. By the first complem. condition, we must have: $\lambda_1 = 0$. Therefore, $\boldsymbol{\lambda} = \mathbf{0}$.

- We plug in $r = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$ back into KKT conditions. It is easy to conclude that all $(0; h)$ with $h \geq 0$ are KKT points.

**Case 2**: $r > 0$. By second complem. condition, we have $\lambda_2 = 0$.
**Subcase 2.1**: $\lambda_1 = 0$. By the second main condition, we have $\lambda_3 = -\pi r^2 < 0$, so there is no such KKT points.
**Subcase 2.2**: $\lambda_1 > 0$. We have $h \neq 0$ (otherwise the first main condition yields $4\pi r\lambda_1 = 0$). Also, by third complem. condition, we have $\lambda_3 = 0$.

# Example III: Continued

Therefore, in Case 2, $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$.

# Example III: Continued

Therefore, in Case 2, $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$. Then the second main condition implies $\lambda_1 = r/2$. Plug it back in the first main condition, we get $h = 2r$. Because $\lambda_1 > 0$, by the first complem. condition, we have $2\pi r^2 + 2\pi rh = C$.

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Therefore, in Case 2, $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$. Then the second main condition implies $\lambda_1 = r/2$. Plug it back in the first main condition, we get $h = 2r$. Because $\lambda_1 > 0$, by the first complem. condition, we have $2\pi r^2 + 2\pi rh = C$. Combined with $h = 2r$, we solve $r = \sqrt{C/6\pi}, h = 2\sqrt{C/6\pi}$ . And $\lambda_1 = r/2 = \sqrt{C/24\pi}$.

# Example III: Continued

Therefore, in Case 2, $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$. Then the second main condition implies $\lambda_1 = r/2$. Plug it back in the first main condition, we get $h = 2r$. Because $\lambda_1 > 0$, by the first complem. condition, we have $2\pi r^2 + 2\pi rh = C$. Combined with $h = 2r$, we solve $r = \sqrt{C/6\pi}, h = 2\sqrt{C/6\pi}$ . And $\lambda_1 = r/2 = \sqrt{C/24\pi}$.

**To conclude**, all KKT points for this problem are $\{(0; h) : h \geq 0\}$ and $(\sqrt{C/6\pi}; 2\sqrt{C/6\pi})$.

Today's lecture

- KKT Conditions

Next lecture

- More about KKT Conditions
- Convex Functions

Questions?